Sandipan Nath

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Nationality: Indian · **Work Permit:** Germany

About Me

Computer Scientist with proven research experience in the nonlinear dynamics of spiking neural networks, reservoir computing, and reinforcement learning. Master's candidate in Data Science at Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) with an academic foundation from Canada. Incoming Doctoral Researcher at Technische Universität Graz, specializing in the intersection of efficient, brain-inspired machine learning and deep learning.

Education

Technische Universität Graz — Graz, Austria
Ph.D. Candidate in Computer Science (Incoming) — *Expected Start: 09/2026*

- Non-Teaching research assistant position focusing on integration of Spiking Neural Network (SNN) models into wireless communication systems.
- Research Focus: Neuromorphic Computing, Spiking Neural Networks, Efficient AI.

Friedrich-Alexander-Universität Erlangen-Nürnberg — Erlangen, Germany
Master of Science in Data Science — *01/04/2024 – Current*

- Master's Thesis: *Effect of time-delayed coupling heterogeneity on stochastic resonance in neural networks* (Supervised by Prof. Dr. Marius Yamakou).
- Current Grade: 1.3
- Scale: Best 1.0, Worst 5.0

University of Alberta — Edmonton, Canada
Bachelor of Science with Specialization in Computing Science — *04/09/2018 – 22/11/2022*

- Final Grade: 3.34
- Scale: Best 4.0, Worst 0.0

Work Experience

Technische Universität Graz — Graz, Austria
Doctoral Researcher (Incoming) — *Expected Start: 09/2026*

- Project staff member at the Institute of Machine Learning and Neural Computation.
- *40 hours/week*

Siemens Energy Global A.G. — Erlangen, Germany
Automation and Data Science Developer (Working Student) — *07/01/2025 – 15/07/2026*

- Designed and implemented a centralized data management system to streamline data flow between various HVDC system design tools.
- Developed an interactive dashboard UI to manage tool interoperability via the database, incorporating user feedback.
- *15 hours/week*

University of Alberta — Edmonton, Canada
University Teaching Assistant — *04/01/2021 – 20/04/2021*

- Supervised practical labs, marked assignments/tests, and provided coursework assistance.
- *8 hours/week*

Technical Skills

Languages: Python, SQL, MATLAB, C++, C, Bash/Linux
Machine Learning & AI: PyTorch, TensorFlow, Keras, Scikit-Learn, Brian, NEST, NESTML, Gymnasium, OpenCV, NLTK, Reinforcement Learning, Reservoir Computing, CNNs, Neural Networks
High-Performance Computing: HPC Clusters, SLURM, Parallel Computing (Multiprocessing/Job Scheduling)
Data & Databases: Pandas, NumPy, MySQL, SQLite, MongoDB, FireBase
Tools & Platforms: Git, Jupyter Notebook, Azure, Matplotlib, Agile SCRUM

Selected Projects

Reinforcement Learning for Control Theory | *Fraunhofer IIS* Completed

- **Key Skills:** Python, Reinforcement Learning, LSTM, Mixture Density Networks (MDN), Control Theory
- Building a world model using LSTM + MDN to capture and predict complex system dynamics.
- Training a controller to balance a ball on an arc (on a cart) based on the learned dynamics.

Forecastability in Chaotic Neural Dynamics | *FAU Erlangen-Nürnberg* Completed

- **Key Skills:** Reservoir Computing, Echo State Networks (ESN), Nonlinear Dynamics
- Built an ESN to model Hindmarsh–Rose neurons in spiking, periodic bursting, and chaotic states.
- Explored the network's ability to forecast nonlinear and chaotic time series effectively.
- Link: github.com/ND-SoSe24-Project/ND-Project

Object Detection Models for Arctic Wildlife | *University of Alberta* Completed

- **Key Skills:** Python, Computer Vision, PyTorch
- Implemented state-of-the-art computer vision models like YOLO for the localization of caribou in camera traps.
- Evaluated performance in Arctic conditions, focusing on accuracy and computational efficiency.
- Link: github.com/Robertboy18/Object-Detection-Arctic-Wildlife

Modelling Drug Effects in Tumor Cell Dynamics | *FAU Erlangen-Nürnberg* Completed

- **Key Skills:** ODE Modeling, Systems Biology, SciPy, Numerical Methods
- Fitted ODE models to describe cell growth, drug availability, and cell killing mechanisms.
- Simulated and predicted tumor growth responses to various drug regimens.
- Link: github.com/snath47/oncosys-M1

Shortest Path in a Train Network | *FAU Erlangen-Nürnberg* Completed

- **Key Skills:** Dijkstra's Algorithm, Graph Theory, Data Structures, Python
- Implemented Dijkstra's algorithm to determine optimal routes using variable cost functions.
- Adapted the system to handle multiple constraints such as time, distance, and transfer frequency.
- Link: github.com/snath47/PathfindingIndianRail

Honours and Awards

Murray Thomas Gibson Memorial Scholarship in Mathematics (University of Alberta) – 2019

University of Alberta Undergraduate Scholarship – 2019

Department of Mathematical Statistical Sciences Scholarship (University of Alberta) – 2018

Language Skills

English: C2 (Proficient/Native level)
Hindi: C2 (Proficient/Native level)
Bengali: Native
German: B1.1 (Intermediate)

Certifications

GRE General Test (ETS): Mar 2024
Verbal Reasoning: 161, Quantitative Reasoning: 162, Analytical Writing: 3.5
***Scales**: Verbal Reasoning: 130-170, Quantitative Reasoning: 130-170, Analytical Writing: 0-6*

TOEFL iBT (ETS): Jan 2024
Total Score 109/120 – Reading: 28/30, Listening: 30/30, Speaking: 23/30, Writing: 28/30

Referees

Prof. Dr. Marius Yamakou
Interim Full Professor (W3)
Department of Data Science (DDS)
Friedrich-Alexander-Universität
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Dr.-Ing. Tobias Feigl
Postdoctoral Scientist
Precise Localization and Data Analytics
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Prof. Dr. Julio Vera-González
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